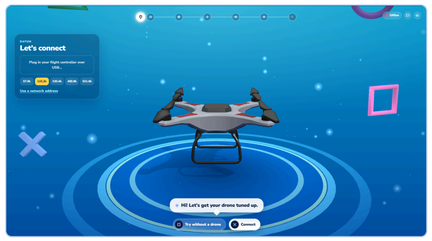
Datum: Interactive Drone Calibration

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1. INTRODUCTION

Before a drone can fly safely, its flight controller must be calibrated: gyroscope, accelerometer, compass, level horizon, radio and ESCs. Ground-control stations offer these procedures as dense forms and text prompts that beginners find hard to follow and easy to get wrong.

Datum turns calibration into a guided, game-like 3D experience inspired by *Astro's Playroom*. A toy drone mirrors the real vehicle in real time, and ghost poses and arrows show what to do next.

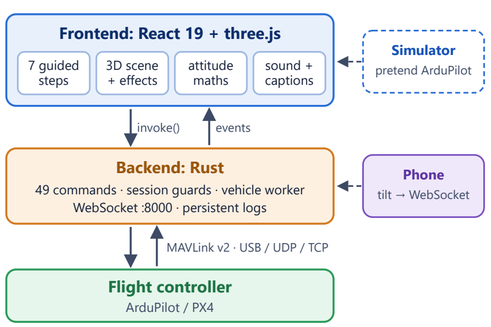


Start screen: seven-step progress track, USB baud picker and simulator mode

2. OBJECTIVES

- Talk **MAVLink** directly to ArduPilot/PX4 over USB, UDP or TCP, with no QGroundControl needed.
- Cover **all six calibrations**: gyro, accel, compass, level, radio, ESC + motor test.
- Make every step **visual**: a live 3D vehicle, pose hints and success screens.
- Offer a **hardware-free simulator** for training and demos.
- Keep it **safe**: bounded motor tests, allow-listed parameters, verified writes.

3. SYSTEM OVERVIEW



Rust backend owns the vehicle; React + three.js owns the 3D experience. Dashed: hardware-free stand-ins.

4. METHODOLOGY

4.1 Guided Calibration Pipeline

Connect (port or simulator) → Gyro (hold still) → Accel (6 poses) → Compass (rotate) → Level (bubble) → Radio (sticks) → ESC (checklist)

- Each step: **Before we start** (board orientation) → **Running** (live 3D guidance) → **Success** or **Failure** with the reason.
- Controller-style input: ✕ confirm, ○ back, □ secondary; short spoken captions.

4.2 MAVLink Command Layer

Step	Command sent to the autopilot	Completion signal
Gyro	`PREFLIGHT_CALIBRATION` p1=1	`COMMAND_ACK`
Accel	p5=1, then `ACCELCAL_VEHICLE_POS` 1–6	success after pose 6
Compass	`DO_START_MAG_CAL` (mask, autosave)	`MAG_CAL_REPORT`
Level	`PREFLIGHT_CALIBRATION` p5=2	`COMMAND_ACK`
Radio	capture `RC_CHANNELS` → `PARAM_SET`	read-back verified
ESC	read `MOT_PWM_TYPE` ; `DO_MOTOR_TEST`	5 ordered steps

- A dedicated worker thread owns the link: 1 Hz heartbeat, message-rate requests, and parsing of every reply into session state.
- Every multi-stage calibration is guarded by a **session ID**; out-of-order steps are rejected.

4.3 Attitude: From NED to the 3D Scene

MAVLink attitude is **NED** (x fwd, y right, z down); the scene is **Y-up**. A fixed permutation C maps vectors, and quaternions are conjugated so rotations still compose:

$$(x, y, z)_{\mathrm{NED}} \mapsto (y, -z, -x)$$

$$q_{\mathrm{scene}} = q_C \cdot q_{\mathrm{NED}} \cdot q_C^{-1}$$

4.4 Compass Coverage Sphere



- **110 cells** on a Fibonacci (golden-angle) sphere.
- Each frame, Earth's field (60° dip) is rotated into the body frame and paints cells within **21°**.
- A hint scores unpainted patches (45° neighbourhood); a cross product gives the axis, shown as "Roll it to the left!"

$$y_i = 1 - 2i/(N-1), \quad r_i = \sqrt{(1 - y_i^2)}, \quad \theta_i = i \cdot \pi(3 - \sqrt{5})$$

4.5 Radio & ESC Logic

Channel travel	Action
≥ 400 µs	**save MIN/MAX/TRIM**
< 100 µs	unused, skipped
100–400 µs	ask to move fully

MOT_PWM_TYPE	Endpoint cal.
PWM / OneShot	**required**
DShot 150–1200	not needed
Motor test	1–15 %, ≤ 5 s

4.6 Technology Stack

Rust · Tauri 2 · mavlink 0.17 · serialport · tokio + axum · React 19 · TypeScript · three.js · react-three-fiber · GSAP · Tailwind 4 · Web Audio

5. RESULTS

5.1 Calibration Outcomes

Step	Visual guidance delivered	Status
Gyroscope	Orientation check, stillness detection, success card	✓ Complete
Accelerometer	Six poses in autopilot order with a ghost drone	✓ Complete
Compass	Int/ext choice, coverage sphere, fitness bar	✓ Complete
Level horizon	Live roll/pitch read-out and 3D bubble level	✓ Complete
Radio	Live FS-i6 transmitter, unused channels skipped	✓ Complete
ESC + motor test	Protocol detection, 5-step checklist, motors A–D	✓ Complete

All flows verified end-to-end in simulator mode; validation on real hardware is in progress.

5.2 Application Walkthrough



Autopilot Rotation board orientation check

Gyroscope "Gyro is wide awake!"

Accelerometer ghost pose: left side



Level Horizon roll +2.3°, pitch +0.1°



Compass Setup external GPS mast



Compass Result fitness bar + reboot hint



Radio live FS-i6 + FS-iA6B



ESC Calibration step 3: throttle up



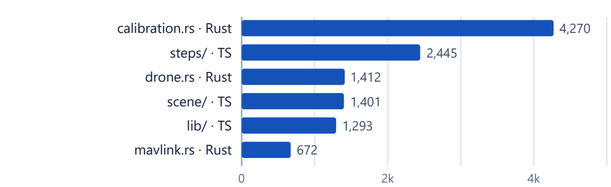
Motor Test spin motor A at 8 %, 2 s

5.3 Automated Tests

Suite	Passed
Backend library	22 / 22
MAVLink monitor	10 / 10
Total	32 / 32

Tests cover command encoding, radio range rules, ESC step ordering, motor-test bounds, the parameter allow-list and loopback-only networking. CI runs them on every push.

5.4 Codebase at a Glance



Source lines by module. About 12.5k lines in total: Rust backend 6.8k, TypeScript frontend 5.7k.

6. DISCUSSION

- Showing the procedure (ghost poses, a filling sphere, a living transmitter) replaces most of the instruction text.
- A single `Backend` interface lets the simulator and real hardware share every screen.
- QGroundControl's forwarding is outbound only, so Datum owns the link directly; only one app can hold the USB port.

Limitations & Future Work

- Validate every flow on real drone hardware.
- Add more 3D transmitter models beyond the FlySky FS-i6.
- Run alongside QGroundControl through a MAVLink router.
- Code-signed installers for Windows and macOS.

7. CONCLUSION

Datum shows that drone calibration doesn't have to be intimidating. It talks MAVLink straight to the flight controller and presents each procedure as a short, visual, game-like sequence. Beginners get the same calibrations as a ground-control station, with far more guidance, while a validated Rust backend keeps the vehicle safe.

Key Takeaways

- ✕ Six calibrations + motor test in one guided flow.
- ○ Live 3D drone driven by MAVLink attitude.
- △ Novel coverage sphere guides compass rotation.
- □ Runs with no drone at all via the simulator.

8. REFERENCES

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9. CONTACT

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